

JAN KWONG

+1 (509) 968-8888 • jankwong705@gmail.com • <https://jankwong705.github.io/me>

EDUCATION

University of California San Diego

La Jolla, CA

M.S. in Computer Science (Anticipated)

Sep 2025 – Dec 2026 (GPA 3.8)

B.S. in Computer Science

Sep 2021 – Jun 2025 (GPA 3.8)

PROJECTS

RAG & Context Engineering Pipeline

La Jolla, CA

Apr 2026 – May 2026

- Built a single-turn Q&A Retrieval-Augmented Generation system over a technical OSS documentation corpus, enforcing a strict 2,000-token per-query context budget
- Conducted a systematic sweep of 8+ configurations (varying chunk size, embedding model, retrieval strategy, and reranker settings) to optimize retrieval and generation quality
- Achieved 0.93 Correctness and 1.00 Faithfulness by implementing parent-document retrieval with BM25/vector hybrid search, a BGE cross-encoder reranker, and a dual-citation few-shot prompting strategy

Security Analysis of Apple Intelligence

La Jolla, CA

Oct 2025 – Dec 2025

- Reverse-engineered macOS binaries using Ghidra to reconstruct the information flow between user input, Generative AI models, and local storage
- Identified a privacy vulnerability where the system persisted unencrypted sensitive personal information to local directories, bypassing the ephemeral-only privacy policy
- Developed a Python forensic script to simulate TCC (Transparency, Consent, and Control) inheritance attacks, successfully exfiltrating sensitive user data from sandboxed environments

Streak-Based Developer Journal

La Jolla, CA

Apr 2024 – Jun 2024

- Collaborated with an 11-member team through Agile practices, participating in regular stand-ups, sprint planning, and conducting retrospectives to ensure smooth and efficient project progress
 - Implemented the tagging system using JavaScript, enabling efficient organization and retrieval of user-generated journal entries
-

EXPERIENCE

UC San Diego

La Jolla, CA

Apr 2026 – Present

Graduate Teaching Assistant

- Leads discussion sections for the class Intro to Research Methods, mentoring students through empirical study design, statistical validation, and analysis methodologies
- Participates in regular instructional meetings to synchronize course operations, address logistical blockers, and align on evaluation standards

UC San Diego LASR Lab

La Jolla, CA

Jan 2024 – Dec 2024

Undergraduate Research Assistant

- Co-authored a research paper submitted to the Acoustical Society of America and PsyArXiv evaluating the performance of OpenAI's Whisper against 75 human transcribers on a dataset of 300 English sentences spoken by 20 speakers with varying accents
 - Developed Python scripts to automate the computation of word error rates (WER), identifying that humans significantly outperform models on isolated words, suggesting potential limitations in models' training data or acoustic context requirements
-

SKILLS

- **Programming Languages:** Python, Java, C, C++, JavaScript, HTML, CSS, SQL, ARM, Clojure, Rocq
- **Libraries & Tools:** Pandas, NumPy, PyTorch, Scikit-learn, Git, Kubernetes, Docker, Ghidra
- **Concepts:** Data Structures, Algorithms, Agile, CI/CD, Machine Learning, NLP, Computer Vision, Data Engineering, Data Analytics, Database Systems, Operating Systems, Computer Architecture, Computer Security
- **Languages:** Native/bilingual proficiency in Cantonese, English, and Mandarin; intermediate in Japanese